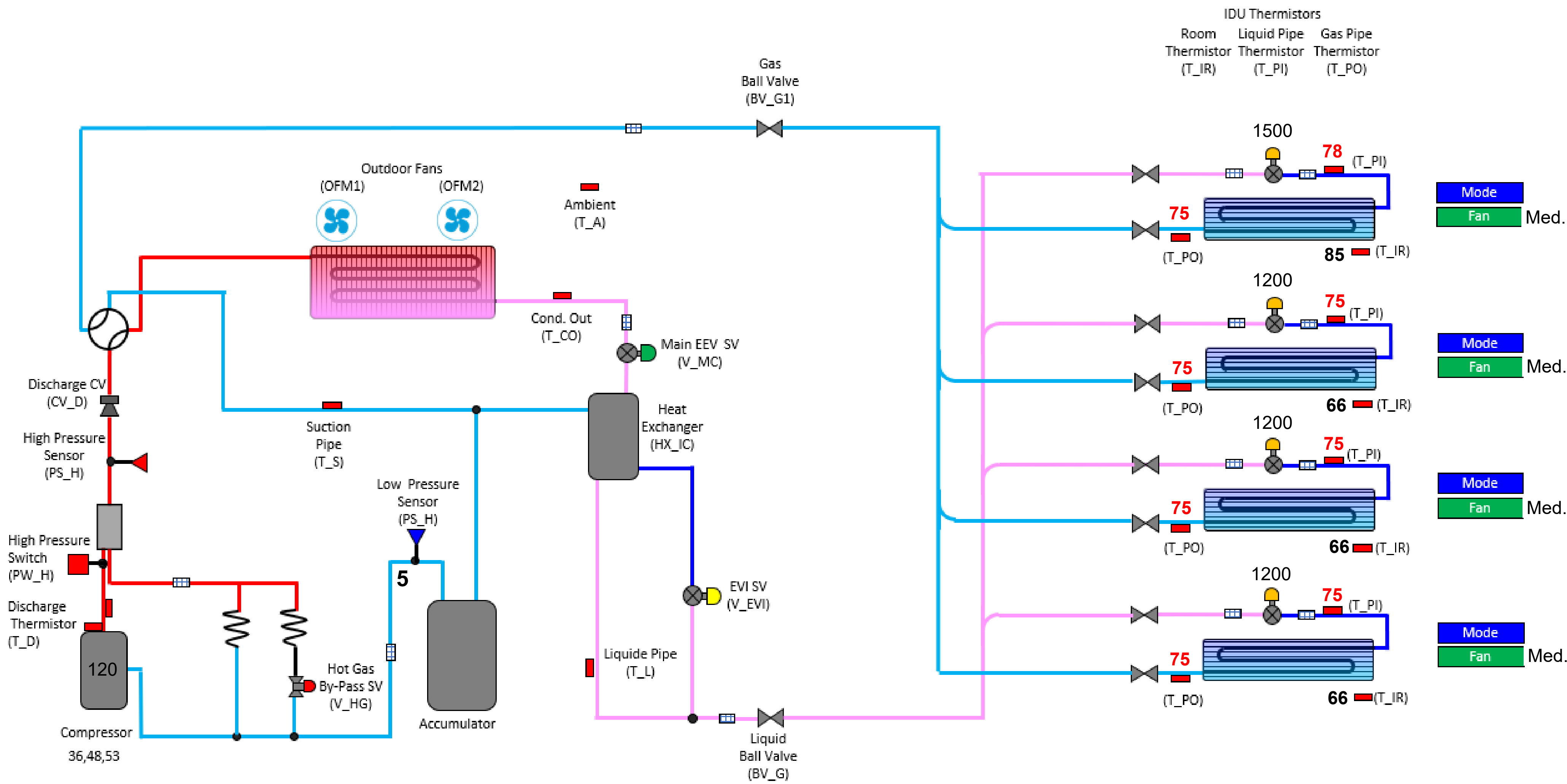
DVM S ECO Heat Pump Operation Evaluation
Cooling Mode



Comments:

Outdoor Unit			
Component	**Target**	**Reading**	
Ambient Temp.			
Compressor HZ	14 - 140	120	Ok
High pressure	355 psig		Ok
H.P Sat. Temp	108^0		
Discharge Temp.	150^0-212^0		Ok
IPM Temp,	Below 194^0		Ok
Low pressure	114 psig	5	Low
L.P. Sat. Temp	38^0		
Suction 1 Temp.			Ok
Suction 2 Temp	N.A.		
S.H.	3.6^0 or 12.6^0	***0***	Ok
Min EEV Position			Ok
Cond. Out Temp.	OAT + (5^0 – 36^0)		Ok
Liquid Line Temp.			Ok
Sub-Cooling @ T_L	9^0 - 36^0	***0***	Ok

Indoor Units			
Component	**Target**	**Reading**	
IDU 1 EEV	250 – 1400 Steps	1500	Ok
IDU 1 SH	3.6^0 (Typical 0^0-7^0)	***-3***	Ok
IDU 1 T PI		78	Ok
IDU 1 T PO		75	Ok
IDU 2 EEV	250 – 1400 Steps	1200	Ok
IDU 2 SH	3.6^0 (Typical 0^0-7^0)	***0***	Ok
IDU 2 T PI		75	Ok
IDU 2 T PO		75	Ok
IDU 3 EEV	250 – 1400 Steps	1200	Ok
IDU 3 SH	3.6^0 (Typical 0^0-7^0)	***0***	Ok
IDU 3 T PI		75	Ok
IDU 3 T PO		75	Ok
IDU 4 EEV	250 – 1400 Steps	1200	Ok
IDU 4 SH	3.6^0 (Typical 0^0-7^0)	***0***	Ok
IDU 4 T PI		**75**	Ok
IDU 4 T PO		**75**	Ok

Directions:
Enter the data in the spaces provided. The reading will populate in the refrigerant diagram. Space temperature and EEV steps are entered at the IDU. Use the drop down boxes to identify the state of the component. Use the operating data and the target operation data on the next page to troubleshoot the system.

DVM S Eco Heat Pump
Target Operation

Air Source HP Outdoor Unit				
Component	**Abbreviation**		**Cooling Mode**	
Compressor	Comp	Operation	Target low pressure control (default: 113.8psig)	Primary function, Maintain Average Evaporator Temperature
		Range	14~140Hz	
Outdoor Fan Motor	OFM	Operation	Target high pressure control (default: 355.6psig)	Fans speed control is used to maintain head pressure
		Range	0-39 step (0-1300rpm)	Typical Head Pressure range 355 psig - 469 psig
Main EEV	E_M	Operation	Full Open	
		Range	2000 Steps	
EVI EEV	E_EVI	Operation	*Subcooling control (default: SC=36°F)	SC = High Side Saturation Temp. - Liquid Line Temp.
		Range	0-480step	
4 Way Valve	V_4W	Operation	Off	
EVI By-Pass Solenoid Valve	V_EB	Operation	On (Open): *T_a > 100.4°F or all compressors in unit are at 50Hz↓	Activates Sub-Cooling Control
			Off (Close): T_a < 97°F and all compressors in unit are at 70Hz↑	Discharge Temperature Protection, Control by unit
EVI Solenoid Valve	V_EVI	Operation	On (Open)	
Hot Gas By-Pass	V_HG	Operation	Off (Closed)	
Accumulator Oil Return Solenoid Valve	V_AR	Operation	On (Open when compressor is running)	**Off** during start up if DSH is < 18℉. **Off** for 2 min. if DSH is < 18℉ **On** when DCSH > 27℉
Hot Gas By-Pass 2 (HR Only)	V_HG2	Operation	Off Closed	Opens after 20 minutes of continuous run time in cooling mode
Main EEV Solenoid Valve (HR Only)	V_ME	Operation	On (Open)	Energized in cooling only
Main Cooling Solenoid	VMC	Operation	Off (Closed)	
Condenser Out Thermistor	T_CO	Value	Should be 5-36 drgrees above outside air temperature	Used to calculate condenser sub-cooling
Component	**Abbreviation**		**Cooling Mode**	**Comments**
Room Temperature	Room Temp.	Operation	65~86℉	
Liquid Pipe Thermistor	EVA In 1	Operation	Evaporator Temperature	Used to calculate Evaporator Superheat
Gas Pipe Thermistor	EVA Out 1	Operation	Measures Suction Pipe Temperature	Used to calculate Evaporator Superheat
EEV	EEV 1	Operation	Modulates to maintain Superheat of 3.6 degrees F	Superheat = Pipe out Temp - Pipe in Temp. (Typical Range 0-7)
		Range	0 - 2000 Steps (0 Steps when zone is satisfied)	Typical range between 250 and 1500 Steps
IDU Fan	Fan Speed	Range	Low , Medium, High, Auto	
		Operation	Low Speed: |Ts - Ti| ≤ 1.8°F	TS - Set Temperature
			Medium Speed: 1.8°F < |Ts - Ti| ≤ 5.4°F	Ti Indoor Temperature
			High Speed 5.4°F < |Ts - Ti|	